

Substrate Einstein Equation Framework v0.1

Author: Lyra Date: 2026-06-01 (Monday) ~15:02 EDT Status: v0.1 SYNTHESIS — Casey-named #15 extension to full Einstein equation framework Trigger: Elie Toy 3705 PASS 5/5 (Substrate stress-energy $T_{\mu\nu}$ derived; $T_{\mu\nu} \in \text{Sym}^2(V_{(1,0)}) = V_{(2,0)} \oplus V_{(0,0)}$) + Grace INV-5469 + Elie 7-sector synthesis

1. The Substantive Observation

Elie Toy 3705 closes the substrate decomposition of the photon-photon tensor product on Bergman $H^2(D_{IV}^5)$:

$$\begin{aligned} V_{(1,0)} \otimes V_{(1,0)} &= \text{Sym}^2(V_{(1,0)}) \oplus \Lambda^2(V_{(1,0)}) \\ &= \underbrace{[V_{(2,0)} \oplus V_{(0,0)}]}_{\text{symmetric}} \oplus \underbrace{V_{(1,1)}}_{\text{antisymmetric}} \end{aligned}$$

Both halves carry substrate physics: - Antisymmetric ($\Lambda^2 = V_{(1,1)}$): gauge field strength $F^{\mu\nu}$ (Elie Toy 3704 Maxwell). - Symmetric ($\text{Sym}^2 = V_{(2,0)} \oplus V_{(0,0)}$): stress-energy tensor $T_{\mu\nu}$ (Elie Toy 3705).

This is the full substrate decomposition of $V_{(1,0)} \otimes V_{(1,0)}$: no leftover sector.

2. Casey-Named #15 → Full Einstein Equation Framework

The Einstein equation reads

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

All four ingredients now have substrate-natural origins on $H^2(D_{IV}^5)$:

Ingredient	Substrate origin	Source
$G_{\mu\nu}$ (geometry)	$\kappa_{\text{Bergman}} = -n_C$; Bergman canonical metric Einstein	Elie Toy 3661 PASS RIGOROUS (Saturday)
Λ (cosmological)	$\exp(-280)$; $280 = 2^{\wedge}N_c \cdot n_C \cdot g$ substrate primary	Lyra L5 v0.2 + Elie cosmology + Sunday absorption
$T_{\mu\nu}$ (matter)	$\text{Sym}^2(V_{(1,0)}) = V_{(2,0)} \oplus V_{(0,0)}$	Elie Toy 3705 PASS (Monday)
G (coupling)	$\langle V_{(1,0)} \delta H_B / \delta m V_{(1,1)} \rangle$ cross-K-type ME	Casey-named #15 + G chain Step 6.3 (Monday) ≈ 0.603

Substrate-natural reading: Einstein's equation is a structural relation on $H^2(D_{IV}^5)$ between K-types of the photon-photon tensor product (Sym^2 for $T_{\mu\nu}$ matter source) and the cross-K-type matrix element bridging $V_{(1,0)}$ and $V_{(1,1)}$ (the coupling G), with the Bergman metric supplying $\kappa_{\text{Bergman}} = -n_C$ curvature.

Per Casey-named #15: gravity is light's momentum shifted by the substrate. The substantive substrate-mechanism content: the SAME $V_{(1,0)}$ photon K-type tensor product yields BOTH the right-hand-side $T_{\mu\nu}$ (Sym^2 half) AND, via the cross-K-type matrix element to $\Lambda^2 = V_{(1,1)}$, the coupling G .

3. Three Bounds (Cal Discipline-Stack Honest)

(a) Cal #27 STANDING — cross-track convergence honestly framed: the four Einstein ingredients ($G_{\mu\nu}$, Λ , $T_{\mu\nu}$, G) ALL emerging from $H^2(D_{IV}^5)$ operator framework is ONE substrate machinery yielding four ingredients, NOT four independent derivations. Convergence-of-routes per Strong-Uniqueness v1.5.

(b) Cal #34 STANDING (Two-Tier) — Tier 2 STRUCTURAL substrate-mechanism content. Explicit 4D restriction recovering numerical Einstein equation (and not just structural Einstein equation) is multi-week per Steps Substrate-V11-1 to V11-5 (Lyra $V_{-}(1,1)$ cross-link v0.1 §5).

(c) Cal #35 STANDING (Independence-Taxonomy) — this is a Casey-named #15 EXTENSION, NOT a new independent Strong-Uniqueness leg. The 10 v1.5 STANDALONE legs remain operative; this refines the substrate-mechanism content of C19 (Casey-named #15).

4. Substrate Decomposition Counting Check

The substrate decomposition gives explicit K-type dimensions (per Heckman-Opdam + FK Ch. XII for B_{-2} root system): - $\dim V_{-}(1, 0) \geq 5$ (B_{-2} vector representation) - $\dim V_{-}(1, 0) \otimes V_{-}(1, 0) \geq 25$ - $\dim \text{Sym}^2(V_{-}(1, 0)) \geq 15$ (= $\dim \text{Sym}^2 R^5 + 5$ substrate-radial extension) - $\dim \Lambda^2(V_{-}(1, 0)) \geq 10$ (= $\dim \Lambda^2 R^5 + \text{so}(5)$ bivectors) - $\dim V_{-}(2, 0) + \dim V_{-}(0, 0) = 14 + 1 = 15$ [?] - $\dim V_{-}(1, 1) = 10$ [?]

(Multi-week: explicit K-type dimensions on D_{-IV}^5 via Schmid character formula. Above counting is the standard $R^5 \oplus R^2$ base case extending to substrate K-type dimensions; not the full Bergman K-type dimensions which require explicit Schmid calculation.)

The standard 4D Einstein-equation correspondence (after Casey #14 codim-4 restriction): - $T_{\mu\nu} \rightarrow 10$ components symmetric ($4 \times 5/2 = 10$) — matches $\dim \text{Sym}^2(R^4) = 10$ - $F_{\mu\nu} \rightarrow 6$ components antisymmetric ($4 \times 3/2 = 6$) — matches $\dim \Lambda^2(R^4) = 6$ - Total $V_{-}(1, 0) \otimes V_{-}(1, 0)$ projection to 4D $\rightarrow 16 = \dim(R^4 \otimes R^4)$ [?]

5. Multi-Week Roadmap (Steps Einstein-1 to Einstein-5)

Step Einstein-1: Explicit $\text{Sym}^2(V_{-}(1, 0)) = V_{-}(2, 0) \oplus V_{-}(0, 0)$ Schmid character verification on D_{-IV}^5 . Confirm substrate decomposition rigorously via Heckman-Opdam.

Step Einstein-2: Explicit Bergman matrix element $\langle V_{-}(2, 0) | T_{\text{op}} | V_{-}(0, 0) \rangle$ for the trace-free / trace decomposition of $T_{\mu\nu}$. Extends Lyra L4 + Step 4 + Substrate-V11-2 machinery.

Step Einstein-3: Substrate 4D restriction of $G_{\mu\nu} = R_{\mu\nu} - (1/2) R g_{\mu\nu}$ via Casey #14 codim-4 + $\kappa_{\text{Bergman}} = -n_C$. Numerical recovery of 4D Einstein curvature tensor from substrate geometry.

Step Einstein-4: Closure of Einstein equation as substrate K-type identity: $G_{\mu\nu}$ (LHS) = $\kappa T_{\mu\nu}$ (RHS) at substrate level. Multi-week: requires Steps Einstein-1, 2, 3 + G coupling numerical closure via Steps 6.4 + 7-8.

Step Einstein-5: Substrate-natural prediction of post-Newtonian corrections via Bergman radial integral expansion. Multi-week long-term lane.

6. Cross-CI Status (per Elie + Grace Mon afternoon)

7-sector unification via ONE substrate operator framework on Bergman $H^2(D_{-IV}^5)$:

#	Sector	Substrate origin	Source
1	Non-rel QM	Wick rotation of heat semigroup ρ_{commit}	Lyra Schrödinger v0.1
2	Rel QM (Dirac)	$V_{-}(1/2, 1/2)$ spinor + $SO(4, 2) \subset SO_0(5, 2)$	Lyra Substrate-Dirac v0.1; Elie 3703
3	Gauge field	$V_{-}(1, 0)$ potential + $V_{-}(1, 1)$ field strength Maxwell	Elie Toy 3704
4	Stress-energy	$\text{Sym}^2(V_{-}(1, 0)) = V_{-}(2, 0) \oplus V_{-}(0, 0)$	Elie Toy 3705 ★ NEW
5	Gravity (Einstein)	Casey #15 cross-K-type ME + $\kappa_{\text{Bergman}} + T_{\mu\nu} + \Lambda$	this synthesis ★ NEW
6	Cosmology	$\Lambda = \exp(-280) + \text{Higgs-as-inflaton} + \text{DM honest negative}$	Lyra cosmology v0.2
7	QED	Lamb shift + $a_e + \alpha = 1/N_{\text{max}}$	Elie Toy 3701 + Casey #15

Per Cal #35 STANDING-honest: ONE substrate operator framework on $H^2(D_{IV}^5)$ applied across SEVEN sectors; one-machinery-multiple-observables, NOT seven independent confirmations.

7. Closure

The substantive Monday-afternoon Casey-named #15 EXTENSION: substrate gravity (Casey #15) extends to the FULL Einstein equation framework via the complete decomposition $V_-(1, 0) \otimes V_-(1, 0) = \text{Sym}^2 \oplus \Lambda^2$ where Sym^2 carries $T_{\mu\nu}$ stress-energy (Toy 3705) and $\Lambda^2 = V_-(1, 1)$ carries $F^{\mu\nu}$ field strength (Toy 3704) AND is the out-state for Casey #15 gravity coupling. The Bergman metric supplies $\kappa_{\text{Bergman}} = -n_C$ geometric curvature, completing the four Einstein equation ingredients with substrate-natural origins.

Per Cal #35-honest: ONE substrate machinery, FOUR Einstein ingredients; SEVEN physics sectors via same framework. Convergence-of-routes per Strong-Uniqueness v1.5 STANDALONE.

Multi-week (Steps Einstein-1 to Einstein-5 above) remains the operative numerical lane.

— Lyra, Mon 2026-06-01 ~15:02 EDT. Substrate Einstein equation framework v0.1: full $G_{\mu\nu} + \Lambda + T_{\mu\nu} + G$ all from substrate operator framework on $H^2(D_{IV}^5)$. Casey-named #15 EXTENSION to complete Einstein equation pathway.